

Ping! Prototype (ONGOING)

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A collaborative “emotional programming” game system that uses rhythm, constraint, and group calibration to model emotional regulation and teamwork

Overview & Purpose

Ping! is the later-stage Wellbotics prototype that reframes “robot programming” into a **collaborative synchronization and emotional calibration experience**. Instead of emphasizing spatial navigation (point A → point B), Ping! focuses on how learners can **recognize, interpret, and regulate emotions** through shared rhythm, constrained communication, and group decision-making.

Ping! was explicitly introduced as a **test build with no robots yet**, designed to establish a working learning system first and then later explore which robots (or custom hardware) best embody the system through motion, sound, color, timing, and feedback.

Learning Goal + Emotional Target

Learning goal/User Story (redefined)

As a **5–8-year-old public school student**, alone or in a group, players engage in a facilitator-supported robot-programming game to meet challenges so that **sequencing and debugging** clearly model **recognizing and managing emotions and behaviors**, aligned with **NYS CS K1.CT.6** and **NYS SEL Benchmark 1A.3A** (as described in the team notes and reflected in the design document’s learning framing).

Emotional target (settled)

Connected.

“Connected” functions as both:

- an emotional outcome, and
- a mechanical requirement (stay in sync; support imbalance; re-align together).

Core Learning Loop

Scenario → plus/minus interpretation → majority check → intensity challenge → reveal

Design Philosophy & Framework: Pattern Language (System-level Prototype)

A key outcome of Ping! is the creation of a **pattern language** (inspired by *The Timeless Way of Building*) to guide all future development: visual design, rules, curriculum, facilitation, and robot behavior.

Core pattern themes (as presented in team notes)

- **Harmonic Learning**: precision + connection coexist
- **Differentiator Sensor**: detects imbalance and triggers rebalancing
- **Collective Retuning / Collaborative Calibration**: system pauses to help slower learners catch up
- **Harmonic Throttling**: system adjusts to advanced learners without leaving others behind
- **Systemic Resonance**: robot joins rhythm as a responsive partner
- **Differentiated Rhythm**: time flexes like breath; fairness comes from pacing
- **Elastic Feedback**: feedback meets effort, avoids punishment
- **Reflective Signaling**: reflection functions as “debugging” for emotions and group process

This framework is important because it changes what “success” means: not speed or individual correctness, but **shared alignment**, repair, and meaning-making.

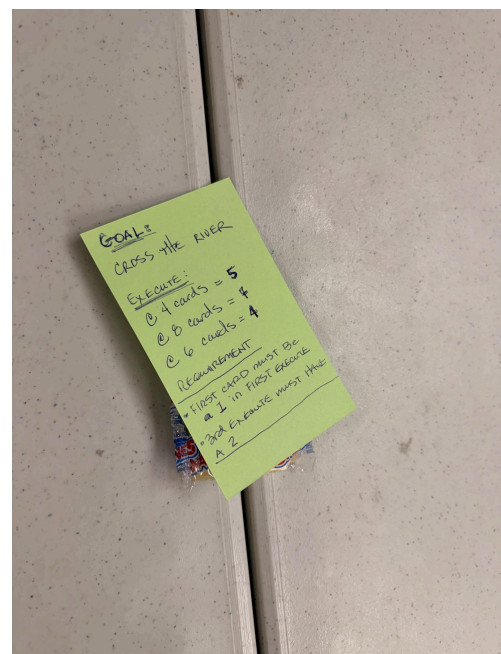
How to Play

Goal of the game

Each team “programs” a robot by building a **hidden sequence of numbered cards** that adds up to the **target total** on the shared **Task Card**, while staying **in sync** (tempo/emotional rhythm) with other teams.

What’s in a team deck (and what the cards do)

- **46 cards per team.**
- **Sequence Cards (20)**: Negafibonacci-ish values (-8, -5, -3, -2, -1, 0, 1, 1, 2, 3, 5, 8), mostly appearing twice; “1” appears four times; includes 2 special declaration cards: **Too Fast** and **Falling Behind**.
- **Dual Emotion Cards (9)**: each has two halves (Happy/Sad, Calm/Worried, Proud/Frustrated, Curious/Confused, Brave/Careful,



Hopeful/Disappointed). Each player sets **one face-up emotion orientation** before play starts.

- **Task Cards (shared deck)**: defines the **target total + card limit** (example “Reach 12 in 6 steps”) for **all teams**.

Setup (step-by-step)

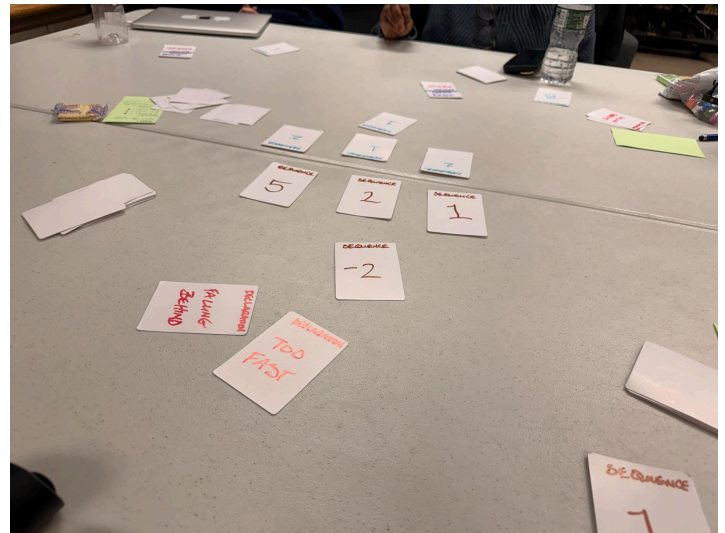
1. Form teams of **2–3 players**.
2. Deal starting hands:
 - 2 players → **3 sequence cards each**
 - 3 players → **2 sequence cards each**
3. Remaining sequence cards become a **face-down draw pile** within reach.
4. Each player gets **3 Emotion Cards**, chooses **1**, places it face-up oriented toward the center.
5. Reveal **1 Task Card** shared by all teams.
6. All teams synchronize by saying “**Ping!**” **three times**; play begins on the third.
7. **All teams play simultaneously.**

Turn structure (the actual “rules of play”)

On a turn, a player chooses **one** of these action types:

A) Program the Robot

- Play **1 sequence card face-down** into your team’s sequence line, left→right.
- No fixed turn order: **any teammate can go next**.
- Constraint: the **far-left card must eventually be a “1”** (teams coordinate to anchor the sequence).
- Stop programming once you’ve placed the **number of cards shown on the Task Card** (e.g., 6), then evaluate.



B) Take an Emotion Action (instead of programming)

1. **Modify Orientation (Self-Support)**
 - Rotate emotion card 180°
 - Discard 1 sequence card → draw 1 new one
2. **Swap a Sequence Card (Blind Peer Support)**
 - Choose a teammate; both secretly pick 1 sequence card and swap face-down
3. **Align & Trade (Visible Co-Regulation)**
 - If your emotion card matches a teammate’s (same card + orientation), announce alignment
 - Trade 1 sequence card **face-up**

C) Play a Declaration Card (Too Fast / Falling Behind)

- **Too Fast:** play if another team/player is surging ahead
- **Falling Behind:** play if your team feels stuck/out of rhythm
- Either one **pauses play**, and all teams re-sync with 3 “**Ping!**” before continuing

Team Beat (rhythm rule)

- After every player on your team has played a card, your team pauses and says “**Ping!**” in unison to help other teams sense your pacing.

Sequence Evaluation (how success/failure works)

After you hit the Task Card’s card limit:

1. Reveal the sequence left→right
2. Add the numbers
3. Compare to target:
 - Exact = robot succeeds; team “syncs”
 - Over/under = robot does alternate response (light/sound/story cue)
 - Negative total = robot “rethinks,” teams collaborate to repair + try again
4. Teams that reach **similar totals** are “in resonance.”

Reflection phase (optional but intended)

- Players can re-orient emotion cards to show how the round felt; facilitator prompts (e.g., “What did the robot show us?”), then draw a new Task Card.

Ending / winning condition (as currently written)

- No single winner.
- Game ends after **three Task Cards [TBD]**.
- If all teams complete at least one task in shared resonance → **Harmonic Win**.

Current Prototype Form: Card-Based “Robotless” Playtest System

Ping!’s most concrete prototype artifact is a **card-based playtest rule set** meant to simulate robot-programming and observe whether the system produces the intended learning and emotional outcomes.

Goal of play

Teams cooperatively play sequence cards to match a **Task Card target total** under constraints of timing, coordination, and emotional alignment. The win condition is **collective**: groups aim for **harmonic completion** rather than beating each other.

Core components (as specified in the PDF)

- **Sequence Cards (Negafibonacci-based)**: includes positive and negative values, with repeated 1s (and special cards such as “Too Fast” / “Falling Behind” included in the system)
- **Dual Emotion Cards**: paired emotions (e.g., Happy/Sad; Calm/Worried; etc.), oriented to represent current state
- **Task Cards**: define numeric target and step limit (e.g., reach X in Y steps)
- **Synchronization ritual (“Ping!” count-in)** and recurring **team-beat** check-ins

Turn actions (core mechanics)

Players can:

1. **Program** by playing a sequence card into a shared line
2. Take an **emotion action** that changes coordination/resources (self-support rotation; blind swap; alignment-based trade)
3. Play **declaration cards** (“Too Fast” / “Falling Behind”) to pause the system and re-sync

Evaluation & feedback

Teams reveal totals and compare to target:

- exact match = success / “sync”
 - over/under/negative = alternate system response and collaborative repair
- Reflection is embedded as short facilitator prompts (treated as emotional debugging).

Scenario System (Ongoing development Focus)

Why “scenarios” became central

Through playtesting and discussion, Ping! increasingly shifted toward using **scenarios** as the core input that drives emotional interpretation. The system became less about “solving a math pattern” and more about:

- interpreting what a situation *suggests emotionally*, and
- calibrating the **intensity** of that emotion together.

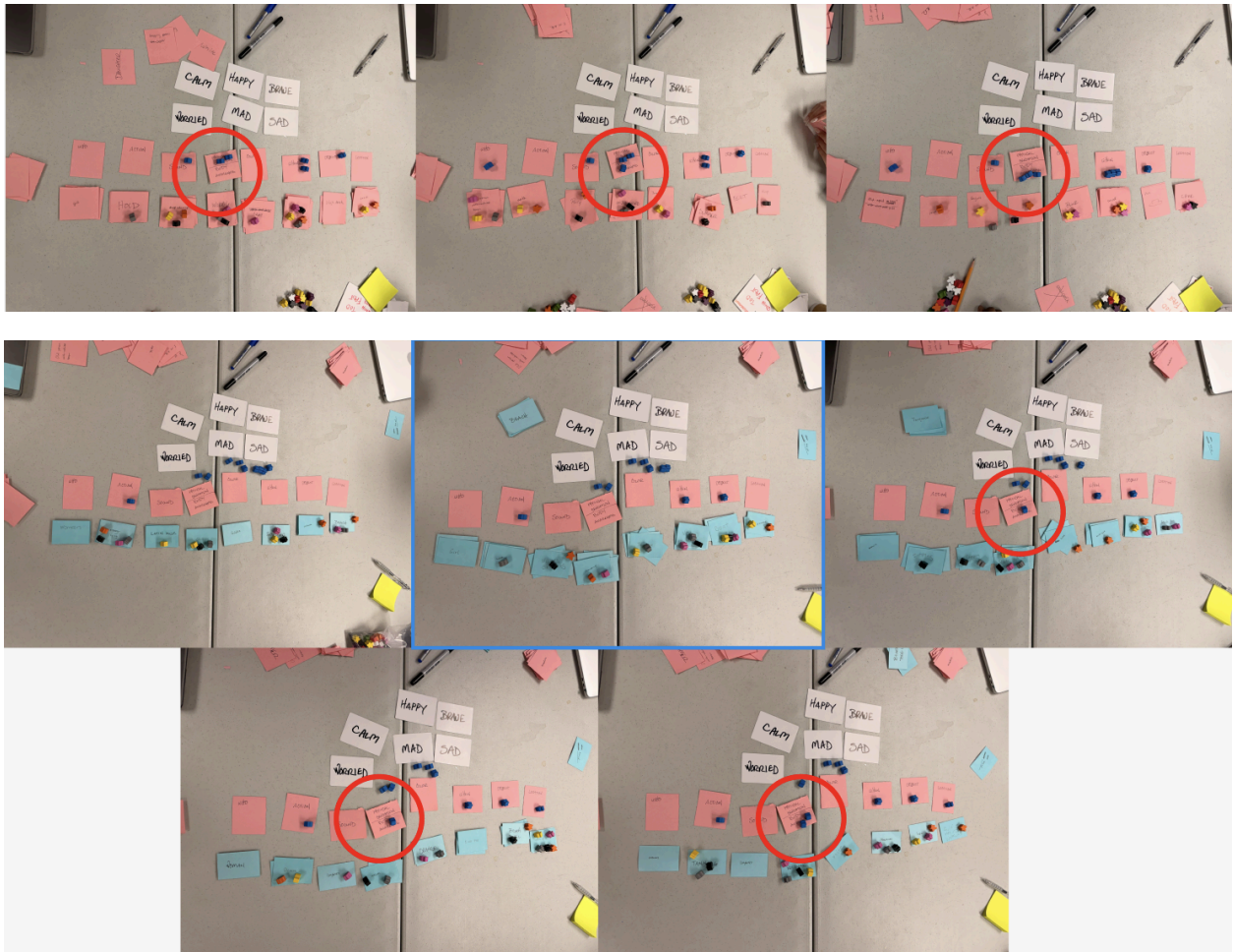
Scenario “category” framework (the newest prototype layer)

The team pressure-tested a set of **eight categories** for describing scenarios:

- **Who** (person/character)
- **Action** (what’s happening)
- **Sound** (auditory elements)
- **Physical sensation / Body awareness** (bodily feelings)
- **Color** (visual descriptors)
- **When** (timing/context)
- **Object** (physical items)
- **Location** (where it occurs)

Key finding / team agreement:

Physical sensation / body awareness consistently affects emotional interpretation the most, and the team agreed it should be a primary lever for scenario design. (This aligns with Wellbotics' emphasis on embodiment.)



Scenario delivery mechanism evolution

Ping! evolved from “cards only” to experimenting with **who/what sets the scenario**:

- Initially: **a human** sets the scenario (live storytelling), which playtests suggested was more engaging and intuitive.
- Explored: **images** as prompts, which felt more systematic/less engaging.
- Current direction: exploring scenarios delivered by **AI** and/or **illustrations/videos**, as a way to:
 - standardize prompts,
 - vary difficulty and context,
 - support scalable facilitation,
 - and keep the system engaging while remaining developmentally appropriate.

Prototype Evolution

Across iterations, Ping! moved through several stages:

Stage 1: Synchronization + constraint language

- Beep/buzz limited communication tested (robot-like language constraint)
- Synchronization rituals and tableau layout helped teams “see” coordination
- Problems discovered: awkward onboarding, pre-planning loopholes, emotion layer not integrated enough, math complexity too high for target age

Stage 2: Emotion guessing as gating mechanism

- One player sets the robot’s emotion from a scenario; others guess
- Correct guesses allow programming actions
- Problems discovered: guessing bottleneck slowed progression; emotion could feel disconnected from the actual collaborative task unless tightly integrated

Stage 3 (breakthrough): Emotions as intensity spectrum

A major shift occurred: Ping! works best when it moves beyond “labeling the correct emotion” into:

- emotions as **degrees** (a lot vs a little),
- group understanding as **comparison and adjustment**, not unanimity.

This became the “locked-in” insight guiding later rule sets (plus/minus intensity modeling).

Current Status & Next Step: Robot Integration

Ping! is explicitly **ongoing**, and the current development direction is to add robot embodiment once the scenario + calibration system is stable. Robot feedback operates at the **system level**, not the personal level, while Correction is framed as something **shifted**, not someone failed.

Planned robot role (direction, not finalized)

Robots will likely map scenario interpretation and intensity into:

- **motion patterns**
- **sound**
- **color/light**
- **timing/rhythm**
- possibly “strain” or “misalignment” behaviors that invite re-tuning

Design Constraints

- Do not collapse intensity back into correctness

- Do not force verbal explanation
- Do not privilege unanimity over coordination
- Do not add complexity that breaks accessibility

Why this matters

Robot feedback will turn abstract emotional reasoning into something learners can:

- observe,
- discuss,
- debug,
- and regulate together.

Conclusion

Ping! represents the most recent Wellbotics prototype shift from “robot navigation tasks” toward an **algorithmic system for emotional communication and co-regulation**. It began as a card-based simulation of robot programming with synchronization constraints and evolved—through iterative playtests—toward a scenario-driven emotional calibration framework where learners compare interpretations, negotiate intensity, and repair misalignment together.

The current focus is refining the **scenario language** (Who/Action/Sound/Body awareness/Color/When/Object/Location), with the team agreeing that **physical sensation/body awareness** is the strongest driver of emotional interpretation. Ping! is now moving from “simple card game prompts” toward scenarios delivered by **human facilitation**, potentially **AI**, and **illustrations/videos**, while planning for a future stage where robots embody group consensus through multimodal feedback (movement, sound, color, rhythm).

This makes Ping! a later-stage prototype not because it is finished, but because it has clarified a distinctive design thesis: **connection is not a theme—it is a system behavior**.

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